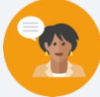


9.4 Interpreting Box Plots

Lesson Introduction

 Benchmark	MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data.		 Learning Targets	- I can determine the minimum, lower quartile, median, upper quartile, and maximum from a box plot. - I can determine the interquartile range from a box plot. - I can describe the spread and distribution of data from a box plot.	
 Benchmark Coherence	Building On N/A		Working Toward MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.		
 Essential Questions	- How do you interpret data from a box plot? - How does a box plot show range, minimum, maximum, median, and quartiles? - How would you find interquartile range from a box plot?				
 Lesson Narrative	This lesson explores how to use a box plot to interpret a set of data. Students construct and use a box plot to answer key questions about data. Students find the range, minimum, maximum, median, and quartiles of a data set as well as use the information to find the interquartile range.				
 Component Summary	9.4.1 Warm-Up (~5 min.)	Examining and discussing similar numbers (<i>SE p. 57</i>)	9.4.4 Guided Practice (5 min.)	Determining interquartile range (<i>SE p. 60</i>)	
	9.4.2 Exploration (10 min.)	Exploring a box plot (<i>SE p. 57</i>)	9.4.5 Try It (5-10 min.)	Using box plots to understand data (<i>SE p. 61</i>)	
	9.4.3 Guided Instruction (10 min.)	Learning about box plots and interquartile range (<i>SE p. 59</i>)	9.4.6 Your Turn! (5 min.)	Practicing working with box plots (<i>SE p. 62</i>)	
	9.4.7 Wrap-Up (~5 min.)	Describing box plots and how to use them (<i>SE p. 63</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Box plot - Quartile - Interquartile range <u>Additional Terms:</u> N/A		(Optional) Chart paper (Optional) Set of data for students to use to create a box plot		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse the median of the data with its mean. - Students may struggle to understand the purposes, placements, and meaning of the quartiles. - Students may confuse the range of a box plot with its interquartile range and how to find each from the data display.	- Highlight how to find the median of the data from a box plot. Prompt students to notice that the mean is not reflected in a box plot. - On chart paper, create a list of how to interpret each part of a box plot and what it means. Display this list for students to refer to throughout the lesson. - Highlight the different values used to calculate range and interquartile range. Prompt students to discuss how and when it would be helpful to use each one in a real-life context.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 9.4.2 Exploration positions students to notice and interpret the details of box plots. Consider applying Notice and Wonder routines for each box plot, positioning students to connect the data values and their representation on the box plot.	MA.K12.MTR.3.1: Mathematical Fluency Students will be working to create and interpret box plots. Clarifications within this standard emphasize accurately completing mathematical tasks and adapting procedures to new contexts. Provide students with the opportunity to practice the processes for interpreting a box plot and to reflect on the processes they use.
 Differentiate Instruction	Box plots offer visual entry points for students as they interpret and analyze data sets. Reference the effectiveness of multiple representations (MA.K12.MTR.2.1) when interacting with the data throughout the lesson. Using colors to label the different sections of the box plots provides an additional visual for students to use to interpret the data and connect the visual representation of the box plot with the data set.	
Lesson Notes:		

9.4.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students examine four numbers—a percentage, a fraction less than 1, a fraction greater than 1, and a repeating decimal—to determine which one is not like the others. Students will then explain why that number differs from the others. Students can accurately answer that any of the numbers does not belong as long as they can support their answer with solid, mathematical reasoning.



9.4.1 Warm-Up: Which One Doesn't Belong?

Examine the four numbers.

33%	$\frac{1}{3}$
$\frac{5}{3}$	$0.\overline{6}$

- Which of these numbers does not belong? Explain your thinking.

Answers will vary.



Challenge students to create additional groups of four numbers in which each number could logically not belong. Have students switch their numbers with another student or small group to discuss.

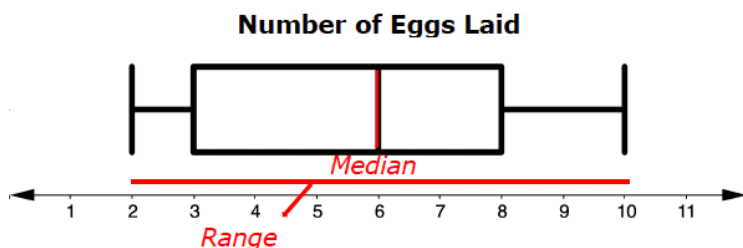
9.4.2 Exploration: Exploring Box Plots

Component Overview: Students will examine a box plot and use it to answer questions about the data expressed in it. Students may work with a partner on the entire activity or in small groups.



9.4.2 Exploration: Exploring Box Plots

1. A farmer recorded the number of eggs her hens laid in one week. The data set was the total number of eggs each of her 53 hens laid during the week. Below is a box plot representing the number of eggs laid by each hen during the week.



With your partner use the box plot to complete the statements.

- a. The smallest number of eggs laid by the hens is 2.
- b. The largest number of eggs laid by the hens is 10.
- c. Find another pair of partners and compare the smallest and largest number of eggs laid. Discuss how you determined the values from the box plot. If your values are different, come to a consensus on the smallest and largest number of eggs laid by the hens, and circle these values on the box plot.

Answers will vary. The smallest value is the first number in the 5-number summary, and in this scenario the smallest value is 2 and the largest value is 10.

Smallest Number of Eggs - 2

Largest Number of Eggs - 10



Provide scaffolded questions to bridge instructional gaps as students explore in groups:

- What is the lowest number in the data set? What does this represent? How is this value shown in the box plot?
- What is the greatest number in the data set? What does this represent? How is this value shown in the box plot?

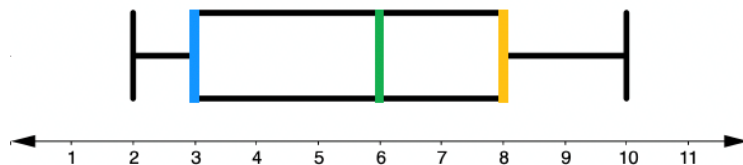


Students may struggle to interpret the box plot, but that is the purpose of the exploration – to provide students an opportunity to explore an unfamiliar data display and use what they know to begin interpreting its meaning. Consider ways to encourage and support your students as they work through the process with their partners or small groups. Consider also the purposeful grouping used for this activity to ensure rich discussions can be added in each group.

9.4.2 Exploration: Exploring Box Plots (continued)

- d. Label the range for the number of eggs laid on the box plot.
Range = Maximum - Minimum = $10 - 2 = 8$
- e. With your partner, conjecture the median number of eggs laid. Label this number as "median" on the box plot.
Labeled on box plot. The median is the middle value of a data set. It is the line in the middle of the box, 6.
- f. The first quartile is highlighted in blue, the second quartile is highlighted in green, and the third quartile is highlighted in yellow.

Number of Eggs Laid



If the second quartile is the median, M , write a conjecture with your partner about how the first and third quartiles are each determined.

Answers will vary. The first quartile is the first quarter of data, so it is the middle of the first half of data. The third quartile is the third quarter of data, so it is the middle of the upper half of the data values when ordered from smallest to largest.

2. Work with your partner to complete the statement.
A **box plot** divides data into 4 equal parts that each contain roughly 25% of data from a given data set.
3. Explain why you chose the value you used to complete the statement.
Answers will vary.



Notice and Wonder

Engage students in a Notice and Wonder routine to begin to analyze and draw conclusions about the value of representing data in box plots. Encourage students to notice similarities between their previous use of range and median and how they are used in box plots. Prompt students to think about why median, not mean, is part of a box plot.

9.4.3 Guided Instruction: Interpreting Box Plots

Component Overview: Students label parts of a box plot and formalize how to interpret it following their exploration. Students use the information in the box plot to determine the interquartile range. The Guided Instruction provides entry points for both visual and auditory learners. This is an opportunity to formalize what students explored in the previous component to ensure understanding of how to interpret data represented in a box plot.



9.4.3 Guided Instruction: Interpreting Box Plots

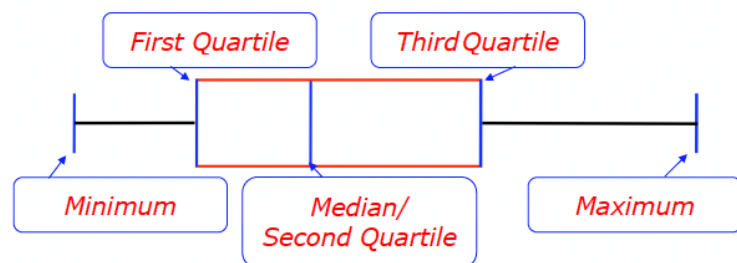


A **box plot** is a plot displaying the spread or distribution of a data set using a five-number summary: the minimum, lower quartile (first quartile), median, upper quartile (third quartile), and maximum. It is also called a box-and-whisker plot.



Consider adding numbers to the box plot in the 9.4.3 Guided Instruction if you think it will make the activity less abstract for students, or refer to the box plot they explored in the 9.4.2 Exploration. Use the data points to help students determine the minimum, maximum, median, quartiles, and interquartile range.

1. Label the maximum, minimum, median, first quartile, and third quartile on the box plot.



2. Complete the statement.

The **first quartile (Q_1)** is the middle value of the data

that are ☐ less than the median.

- ☐ greater than
- ☐ equal to

The **third quartile (Q_3)** is the middle value of the data

that are ☐ less than the median.

- ☐ greater than
- ☐ equal to

9.4.3 Guided Instruction: Interpreting Box Plots (continued)



For a data set with median, M , the first **quartile** is the median of the data values less than M , and the third **quartile** is the median of the data values greater than M . The second **quartile** is the median, M .

3. Complete the statement.

The measure of central tendency used to create a box plot is the _____, which divides the data set in half.

- ☐ mean,
- ☒ **median,**
- ☐ mode,

A number line often accompanies a box plot and is used to determine the values associated with each part of the plot.

Another measure of variability like **range**, the **interquartile range (IQR)** can be found by determining the difference in the third quartile from the first quartile.



A measure of variation in a set of numerical data, the **interquartile range** is the distance between the first and third quartiles of the data set.



$$\text{IQR} = (Q_3) - (Q_1)$$

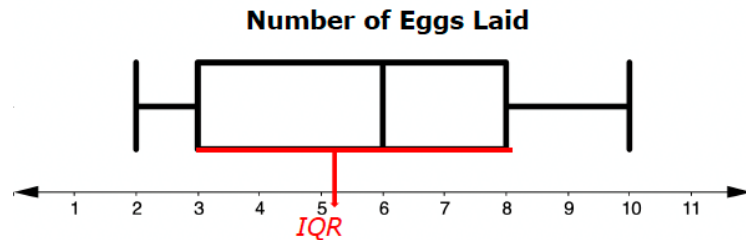


Encourage students to examine range and interquartile range by asking the following questions:

- How do you find the range? What does it represent?
- How do you find the interquartile range? What does it represent?
- How do range and interquartile range differ? How are they similar?

9.4.3 Guided Instruction: Interpreting Box Plots (continued)

4. Revisit the box plot of the eggs laid by a farmer's hens.



- a. What is the interquartile range (IQR) for the box plot of the context?

$$\text{Interquartile Range (IQR)} = Q_3 - Q_1$$

$$\text{Interquartile Range (IQR)} = 8 - 3$$

$$\text{Interquartile Range (IQR)} = 5$$

- b. Label the IQR on the box plot.

Shown on box plot.



Encourage students to think about the term “interquartile” and its relationship to the prefix “inter-,” meaning “between or among.” Discuss that the interquartile range is the range between the range created by the minimum and maximum values and also the range between the first and third quartiles.



Consider using different colors to highlight the different sections of the box plot. Use the same colors to show the calculations for how to arrive at the data points.

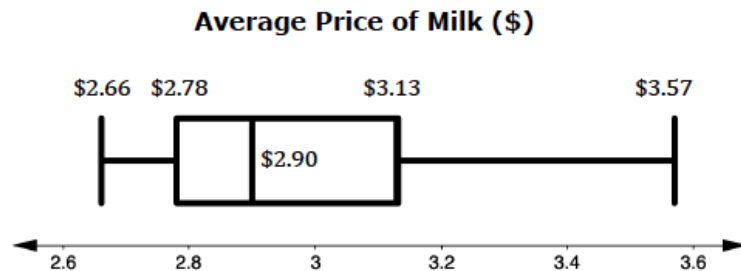
9.4.4 Guided Practice: Interpreting Box Plots

Component Overview: Students determine the range and interquartile range of the data represented by the box plot. Through the box plot, students can examine a real-world example of how to display data.



9.4.4 Guided Practice: Interpreting Box Plots

1. The average price of milk, in dollars (\$), for each month from April 2000 to November 2006 was used to create the box plot shown.



Use this section to focus students on range and interquartile range. If needed, review how to find the minimum, maximum, and quartiles as a vehicle to determining the values of range and interquartile range.

- a. What is the range of the data?
 $3.57 - 2.66 = 0.91$
- b. Determine the interquartile range.
 $3.13 - 2.78 = 0.35$
- c. Does the average cost of milk from April 2000 to November 2006 vary a lot or a little? Explain your reasoning.
Answers will vary. The range of nearly one dollar is almost half of the cost of milk in 2000, so that may be seen as a large variation.

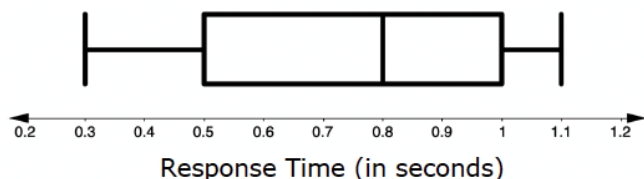
9.4.5 Try It: Interpreting Box Plots

Component Overview: Students can synthesize their findings in this activity. In the Try It, students will be utilizing all the knowledge and skills they have previously learned about box plots.



9.4.5 Try It: Interpreting Box Plots

- The box plot displays data on the response times of 100 mice when seeing a flash of light.



Complete the statements.

- The fastest response time is 0.3 seconds.
 - The slowest response time is 1.1 seconds.
 - The middle response time is 0.8 seconds.
 - The lower quartile is 0.5 seconds.
 - The upper quartile is 1 seconds.
 - The range is 0.8 seconds.
 - The interquartile range is 0.5 seconds.
- With your partner, use the values to write a description of the spread and distribution of response times.
Answers will vary. The mouse that responded the slowest to the flash of light was 1.1 seconds. The fastest response to the flash of light was 0.3 seconds. The range of difference between the fastest and the slowest time was 0.8 seconds. This is a large variation since the range is much larger than the fastest time.



“Fastest” is sometimes confused with “the largest,” so students may incorrectly label the fastest as the maximum instead of the minimum time.



Provide scaffolded questions to encourage students to further examine the box plot:

- What is the range of the data? How would you find it?
- What is the first quartile? How would you find it?
- What is the third quartile? How would you find it?
- How do you use those values to find the interquartile range?

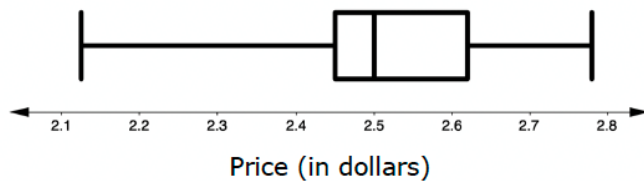


Consider allowing students to work with partners as needed. Partners can work together on each item or divide the work in half and then explain their findings to each other.

9.4.5 Try It: Interpreting Box Plots (continued)

3. The U.S. Energy Information Administration reports weekly average gas prices for the United States. The box plot represents the weekly average gas prices in 2019.

2019 Weekly Gas Prices



What percentage of weeks had an average gas price of at least \$2.50?

50%



Encourage students to more closely examine the box plot:

- Which value represents the highest gas price? How is this represented on the box plot?
- Which value represents the lowest gas price? How is this represented on the box plot?
- What is another name for the middle response time?

9.4.6 Your Turn!

Component Overview: Students continue working with box plots. The additional practice provides students with the opportunity to continue the processes for interpreting a box plot.



9.4.6 Your Turn!

1. Mrs. Henry's 3rd period English class took a spelling test with 12 words on it. A box plot was created from the number of questions students got correct on the test.

Number of Correct Questions



Complete the statements.

- a. The lowest score is 1.
- b. The highest score is 12.
- c. The middle score is 7.
- d. The lower quartile is 2.
- e. The upper quartile is 9.
- f. The range is 11.
- g. The interquartile range is 7.
- h. Write a description of the spread and distribution of the scores.

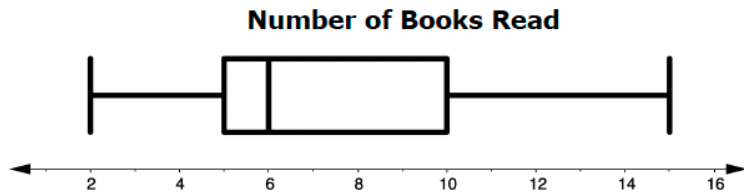
Answers will vary. The student with the lowest number of correct questions in Mrs. Henry's class answered only 1 question correctly and the student with the most correct answers answered all 12 correctly. This is a range of 10 questions. The middle 50% of students answered between 2 and 9 questions correctly, which is the interquartile range of 7 questions. In either case, this shows that the spread of distribution is large, there is large variation in the number of questions answered correctly.



Challenge students to describe the spread and distribution of scores in as much detail as possible. Ask students to think about why the data is distributed as it is and how this may vary in different situations.

9.4.6 Your Turn! (continued)

2. The librarian at Franklin Middle School surveyed students when they returned from summer break on how many books they had read. She created a box plot to represent the results of her survey.



What percentage of students read between:

- 5 and 10 books over the summer? 50%
- 10 and 15 books over the summer? 25%
- 5 and 6 books over the summer? 25%
- 2 and 10 books over the summer? 75%



Consider an extension project of your own classroom or school in which students in groups must collect the data, construct a model, and interpret the data and their findings as they continue working through this unit.

9.4.7 Wrap-Up: Cool Down

Component Overview: Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' ability to construct a box plot and describe its components.



9.4.7 Wrap-Up: Cool Down

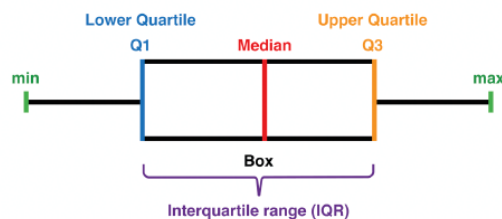
Jamal missed class today and he needs help with what he missed.

- Write a note to Jamal letting him know what a box plot is and what it is used for. Draw a picture or diagram that may help him better understand.

Answers will vary.

Dear Jamal,

You missed a fun class today on box plots. Box plots display the 5-number summary – minimum, first quartile, median/second quartile, third quartile, and maximum. The box plot has four parts that each represent 25% of the data, and it does not matter how big the part is, it still represents 25% of the data values. A box plot can be used to observe the distribution of the data. Here is a picture of a box plot to help:



For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.